Non-stationary simulation study: what a mean-only MRTS augmentation can and cannot buy

July 2026 — interim report v4 (settings 1–12, the threshold-vs-symmetry disambiguation, the $K \geq 30$ extension, and the non-t extension pilots 13–21)

Abstract

We evaluate the MRTS covariate extension of the Morris et al. (2017) skew-t space–time model on twelve simulated data-generating processes (DGPs) that place non-stationarity in different moments: the mean (settings 1, 2, 11, 12), the spatial dependence (settings 3–8), the tails (setting 9), and the skewness (setting 10). Three findings. (i) MRTS improves predictive scores *only* where the truth carries time-averaged mean structure, and this is provable in advance: an oracle calculation shows the relative Brier score has a floor $R^\star \approx 1$ for nine of eleven settings, so their observed nulls are properties of the design, not evidence about MRTS. (ii) Where mean structure exists, the energy- and variogram-score gains are carried *entirely* by the two thresholded symmetric-t methods. A disambiguating pair of methods (symmetric-t *without* threshold, $n = 10$) attributes this cleanly to the *threshold*, not the symmetry — and the absolute scores reframe it: POT censoring without a mean basis damages the predictive field so badly (baseline energy 2.6×–16× worse) that MRTS is a partial *rescue*, not a synergy; the unthresholded methods never needed rescuing and remain better in absolute terms at every K. (iii) The Brier score is close to blind to mean recovery and can even deteriorate with K: in the strong-mean setting 11 a fit that recovers every parameter of the DGP at $K = 25$ has a *worse* Brier score than the badly misspecified $K = 0$ baseline. The rich-mean setting 12, run out to $K = 50$, resolves this: the deterioration is confined to intermediate K (where the mean basis is still too coarse), and once the basis is fine enough ($K \geq 30$–40) the Brier score improves by 7–12%, consistent with the extrapolation-error mechanism. The discriminating check passes: the simple-surface setting 11, extended to the same $K = 50$, never turns positive — its Brier ratio only decays back to parity ($1.09 \to 1.00$) — so the reversal requires mean structure that only a fine basis can resolve, not merely a large K. New in v4: the nine non-t extension settings (13–21; Gaussian-GP errors, so the skew-t error law is misspecified) are fitted at pilot scale ($n = 3$) and included in the score tables; their analysis is deferred.

1 Design

Data-generating process. Settings 1–12 share the skew-t-1 core on $n_s = 144$ sites drawn uniformly on $[0, 10]^2$ (settings 13–21 replace it with a Gaussian process; see below):

$$Y_t(\boldsymbol{s}) \;=\; 10 \;+\; \lambda\,\sigma_t\,|z_t| \;+\; \sigma_t\,v_t(\boldsymbol{s}) \;+\; [\text{structure}]_k, \qquad t = 1, \ldots, 50, \tag{1}$$

with $\lambda = 3$, $z_t \overset{iid}{\sim} N(0,1)$ half-normal, $v_t \sim N(0, C)$ exponential correlation ($\rho = 1$, $\gamma = 0.9$), $\tau_t \sim \mathrm{Ga}(3/2, 4)$, $\sigma_t = \tau_t^{-1/2}$ (marginal t_3), i.i.d. in time. The settings differ only in the structure term—or, for the *replace-C* settings, in the error correlation itself (Table 1). Table 2 gives the resulting anatomy of each DGP: the marginal law of $Y_t(\boldsymbol{s})$, the spatial dependence of the error process, and the time-averaged true mean surface $\bar{\mu}(\boldsymbol{s})$ (the target of any pooled-β mean basis). Settings 13–21 form a *non-t extension*: they drop the skew-t generator entirely in favour of a nonlinear mean surface (sd 11) with Gaussian-GP errors ($\sigma_g = 4$, same C_{exp}), so that the mean channel is tested without the crutch of a correctly specified error law at large K. All nine are fitted and scored at pilot scale ($n = 3$, $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$) and appear in the score tables; Section 5 introduces four of them in detail, and the systematic analysis is deferred to the next revision.

Table 1: The DGP settings and the number of simulation replicates n (independent datasets) fitted per setting. Settings 1–12 share the skew-t core; settings 13–21 are the non-t extension (Gaussian-GP errors, $\sigma_g = 4$; nonlinear mean surface orthogonalised against $[1, s_1, s_2]$ and standardised to sd = 11). Routes: *mean* = added to the fixed mean; *additive* = mean-zero random field added to the draw; *replace C* = non-stationary correlation handed to the generator (unit diagonal, so the *marginal* skew-t law is unchanged — verified by KS tests, $p \in [0.075, 0.864]$); *skew term* = spatially varying λ.

ID	`surf_type`	non-stationarity	route	n
1	invariant	mean, 2 cosine bumps, sd = 3.7	mean	10
2	varying	mean, time-varying weights	mean	10[a]
3	ns_dependence	dependence, low-rank cosine RE	additive	10
4	ps_cov	dependence, Paciorek–Schervish	replace C	3
5	ps_add	dependence, same PS field	additive	3
6	covdep_cov	dependence, $\rho(\boldsymbol{s}) = e^{0.7 f_1(\boldsymbol{s})}$	replace C	3
7	covdep_add	dependence, same field	additive	3
8	fuentes_cov	dependence, 4-regime mixture	replace C	3
9	gh_marginal	tails, Tukey g-and-h, $g(\boldsymbol{s}), h(\boldsymbol{s})$	additive	3
10	lambda_varying	skewness, $\lambda(\boldsymbol{s}) = 3 + 3 f_2(\boldsymbol{s})$	skew term	3
11	invariant_strong	mean, 3× setting 1, sd = 11	mean	3
12	mrts_multibump	mean, 6 Gaussian bumps, sd = 11	mean	3
Non-t extension (Gaussian-GP errors; skew-t generator dropped)[b]				
13	franke	mean, Franke (multiscale)	mean	3
14	ridge_curved	mean, curved Gaussian ridge	mean	3
15	annulus	mean, Ricker ring (r_0=3, w=1.5)	mean	3
16	sigmoid_front	mean, tanh arc front	mean	3
17	gp_mean	mean, GP draw (per dataset, fixed in t)	mean	3
18	franke_sas	mean = setting 13; sinh-arcsinh error	mean	3
19	poly2	mean, bowl + saddle (2nd-order poly)	mean	3
20	transform_mix	mean, log + exp-decay + ratio terms	mean	3
21	nonsmooth	mean, hinge + $\lvert\cdot\rvert$ (C^0 creases)	mean	3

[a] setting 2 was fitted at $K = 0$ only (no MRTS sweep); it is excluded from the score tables. [b] settings 13–21 use the grid $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$; all nine are fitted and scored ($n = 3$).

Fitted methods and grid. Five baseline methods are fitted to every dataset: (1) Gaussian; (2) skew-t, $K_{\text{knots}} = 1$; (3) symmetric-t, $K_{\text{knots}} = 1$, POT threshold at $q(0.80)$; (4) skew-t, $K_{\text{knots}} = 5$; (5) symmetric-t, $K_{\text{knots}} = 5$, threshold $q(0.80)$. Each is run across MRTS design widths $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$, where $K_{\text{MRTS}} = 0$ denotes the baseline design $[1, s_1, s_2]$ and $K_{\text{MRTS}} = K$ replaces the design with the K-column `autoFRK::mrts` basis built on the *training* sites only (column 1 = intercept, 2–3 linear, 4:K thin-plate-spline eigenfunctions; $K = 3$ spans the baseline). MCMC: 20,000 iterations, 10,000 burn-in. The random seed is shared across K_{MRTS} within each (method, dataset), so all K-comparisons are *paired*. The $K_{\text{MRTS}} \in \{30, 40, 50\}$ extension is now complete for every fitted skew-t setting (1, 3–12), so the full 12-point grid $\{0, 3, \ldots, 50\}$ underlies all their tables.

Scoring. The last 44 of the 144 sites are held out entirely; all scores are computed out-of-sample on those sites. With $\hat{F}$ the posterior predictive law:

$$\mathrm{BS} = \tfrac{1}{11} \sum_{p} \tfrac{1}{44 \cdot 50} \sum_{i,t} \big(\hat{P}(Y_t(\boldsymbol{s}_i^*) > u_p) - \mathbb{1}\{y_t(\boldsymbol{s}_i^*) > u_p\}\big)^2, \quad p \in \{0.90, \ldots, 0.995\}, \tag{2}$$

$$\mathrm{ES} = \tfrac{1}{50} \sum_{t} \Big[\mathbb{E}\,\|\boldsymbol{X}_t - \boldsymbol{y}_t\| - \tfrac{1}{2}\mathbb{E}\,\|\boldsymbol{X}_t - \boldsymbol{X}_t'\|\Big], \qquad \boldsymbol{X}_t, \boldsymbol{X}_t' \sim \hat{F}_t \text{ (44-dim)}, \tag{3}$$

$$\mathrm{VS} = \tfrac{1}{50} \sum_{t} \sum_{i<j} \big(|y_{it} - y_{jt}|^{1/2} - \mathbb{E}\,|X_{it} - X_{jt}|^{1/2}\big)^2. \tag{4}$$

The Brier score is a per-cell *marginal* score and cannot see spatial dependence; the energy score is the multivariate CRPS (reduces to CRPS at $d = 1$) and is location-sensitive; the variogram score ($p = 1/2$) targets pairwise dependence. Every result below is the *relative* score $R_{m,K,d} = S_{m,K,d}/S_{m,0,d}$ against the same method's own $K = 0$ fit, averaged over datasets; $R < 1$ means MRTS helps.

2 Results

Tables 3–5 report, per setting and method, the best (smallest) $\bar{R}_{m,K}$ over $K > 0$, with the minimising K in parentheses. Asterisks mark $|\bar{R} - 1| > 2\,\widehat{\mathrm{SE}}$ (SE over datasets; with $n = 3$ this is indicative only). Table 6 gives the full setting $\times$ K profile of the relative Brier score, method by method.

2.1 Finding 1: gains appear only where the truth has mean structure — and this is provable a priori

The only large entries in any table are settings 1, 11 and 12 (mean structure), methods 3 and 5: energy and variogram reductions of 34–42% (setting 1), 73–80% (setting 11) and 58% (setting 12). Every dependence, tails and skewness setting is flat to within a few percent, in *all three* scores.

This dichotomy can be established without MCMC. Because the MRTS coefficient β is pooled over time, the most any mean basis can deliver is the time-averaged true mean $\bar{\mu}(\boldsymbol{s})$. Define the oracle relative Brier

$$R^\star \;=\; \frac{\mathrm{BS}\big(\bar{\mu}\big)}{\mathrm{BS}\big(\text{spatial constant}\big)}, \tag{5}$$

both evaluated under the *true* skew-t predictive law. $R^\star$ is a floor: no K, sample size or sampler can push the relative Brier below it. Table 9 shows $R^\star \approx 1$ for nine of eleven settings: their Brier nulls are design properties, not evidence about MRTS. Setting 10 is the striking case: $R^\star = 1.12 > 1$ — supplying the *true* mean makes the Brier score *worse*, because the mean shift induced by $\lambda(\boldsymbol{s})$ is a symptom of a skewness change that the constant-λ model cannot represent; correcting the symptom while the disease remains actively misleads the tail probabilities.

2.2 Finding 2: the gains are carried entirely by the thresholded methods

In Table 3, setting 11 (mean sd = 11), the two POT-thresholded symmetric-t methods reduce the energy score by 73–80%, while the Gaussian and both skew-t methods stay within 2% of their $K = 0$

baselines — despite those same skew-t fits recovering the mean surface almost exactly (recovery RMSE 14.0 $\to$ 0.66 for method 2 at $K = 25$) and recovering every DGP parameter (Section 2.3). Table 10 shows the full K profile.

Methods 3/5 are simultaneously (a) POT-thresholded and (b) symmetric-t, so a disambiguating pair was added: methods 11/12, *identical to methods 3/5 except* threshold = 0, fitted on setting 1 with $n = 10$ (Table 11). The verdict is unambiguous: the unthresholded symmetric-t methods are *flat* (energy ratio 0.99 at every K, indistinguishable from the skew-t reference), while their thresholded twins gain 35–40%. **The threshold, not the symmetry, is the driver.**

The absolute column reframes the finding. The thresholded baselines are *catastrophically* worse to begin with: censoring everything below $q(0.80)$ leaves the constant-mean fit with almost no information about the spatial level of the field, and its out-of-sample predictive field degrades by a factor 2.6 (K1) to 16 (K5) in energy score. MRTS repairs the mean that the censored likelihood cannot learn — a partial *rescue*, not a synergy: even at the best K, method 3 reaches $21.8 \times 0.579 \approx 12.6$, still well above the 8.5 that methods 2/11 achieve *without any MRTS at all*. In absolute terms the unthresholded methods win at every K; the correct reading of Finding 2 is therefore "*only the thresholded methods have room to be helped*," and the practical implication is for POT-style threshold models specifically: if the mean is not modelled well, censored fitting amplifies the damage, and a mean basis buys back part of it.

2.3 Finding 3: the Brier score is nearly blind to mean recovery, and can punish it

Setting 11, method 2 (skew-t K1), dataset 1, posterior means (truth: $\lambda = 3$, $\mathbb{E}[\sigma] = 2.26$, $\rho = 1$, $\gamma = 0.9$):

K_{MRTS}	recovery RMSE	$\hat{\lambda}$	$\mathbb{E}[\hat{\sigma}]$	$\hat{\rho}$	$\hat{\gamma}$
0	14.01	0.15	11.16	2.04	0.99
10	2.51	2.04	3.01	1.21	0.91
25	**0.66**	**3.18**	**1.94**	**0.96**	**0.94**

At $K = 0$ the model cannot represent the sd-11 mean surface and absorbs the spatial variance into an inflated σ (5× truth) with λ crushed to zero. At $K = 25$ *every* parameter is recovered. Yet the $K = 25$ Brier score is *worse* than $K = 0$ by 8.6% (Table 10), and the same reversal holds for methods 1, 4, 5. A correctly specified model loses to a badly misspecified one on this score. The working hypothesis — not yet verified — is an out-of-sample interaction: the sharpened predictive distribution ($\hat{\sigma}$ correct instead of inflated) exposes the residual error of the MRTS mean *extrapolated* to the held-out sites, whereas the misspecified $K = 0$ fit hides that error inside an accidentally conservative (inflated-σ) predictive distribution.

Setting 12 resolves the anomaly. The rich-mean setting 12 (six bumps, sd 11) was run out to $K_{\mathrm{MRTS}} = 50$, where the noise-free MRTS projection error of its surface finally becomes small (0.71 at $K = 5$, 0.13 at $K = 25$, 0.057 at $K = 50$). The Brier profile follows exactly the pattern the extrapolation mechanism predicts: flat-to-worse at intermediate K (ratios 1.00–1.05 for $K \leq 25$, where the basis is still too coarse and the residual mean error is exposed by the sharpened predictive), then *improving* once the basis is fine enough:

relative Brier, setting 12	K=15	K=25	K=30	K=50	recovery at K=50
Gaussian	1.019	1.045	0.988	**0.930**	0.512
Skew-t K1	1.022	1.020	0.980	**0.925**	0.073
Sym-t K1 thr.	0.899	0.918	0.892	**0.880**	0.422
Skew-t K5	0.976	1.022	0.965	**0.913**	0.268

(Method 5 is excluded: its exponential-fallback instability blows up one of the three datasets at large K, ratio 2.7.) The K=50 gains of 7–12% are real but far above the oracle floor $R^\star = 0.495$ — the mean channel is recovered (skew-t K1 recovery ratio 0.073), yet most of the theoretically available Brier headroom is spent compensating the sharpness–extrapolation trade-off. Practically: *for judging mean-structure recovery, the Brier score remains the wrong instrument — it responds only when the basis is fine enough to make the residual extrapolation error negligible; energy/variogram scores (or the recovery RMSE directly) are the informative ones.*

The large-K limit discriminates the mechanism. If the reversal were simply a matter of "K large enough," setting 11 — whose 2-bump surface the basis captures by $K \approx 10$ — should also turn positive by $K = 50$. It does not (Table 10, rightmost columns): the mid-K deterioration decays monotonically (1.16 at K=12 to 1.01–1.00 at K=50 for the skew-t and Gaussian methods) but never crosses below parity, leaving the oracle headroom $R^\star = 0.439$ entirely unused. The weak-signal setting 1 behaves the same way at large K (skew-t K1: 0.985 at K=25, 0.978 at K=50 — smooth, small, no regime change). Same K grid, same machinery, opposite outcome to setting 12: the Brier reversal requires mean structure that *only* a fine basis can resolve; extra columns on an already-resolved surface buy nothing and merely let the sharpness–extrapolation penalty fade.

3 Caveats

- Settings 4–11 are pilots with $n = 3$; settings 1 and 3 have $n = 10$. Starred cells with $n = 3$ should be read as indicative. The small-n *best-over-K* statistic also has a selection bias toward values < 1; flat profiles (e.g. Sym-t K5 rows in dependence settings, best $\approx$ 0.72–0.85 with non-monotone K profiles) are consistent with noise plus this selection effect.

- Method 5 (Sym-t K5 T) occasionally triggers its exponential-covariance fallback and produces isolated outlier fits (e.g. setting 4, $K = 25$: energy ratio 15.8 in one dataset). Best-over-K cells are robust to this; per-K curves at $n = 3$ are not.

- All K comparisons are paired (shared seeds across K), so within-method ratios are considerably more precise than the between-method comparisons.

- The recovery RMSE (not tabulated here) has a known bimodality on flat-mean settings caused by the intercept–λ ridge; see `nonsta_settings_design.md`.

4 The setting-12 showcase: why a large K is worth having

Setting 12's six-bump surface was designed so that the MRTS approximation improves *gradually* with K (noise-free projection error 0.71 at K=5, 0.19 at K=15, 0.057 at K=50), in contrast to the 2-bump settings which the basis captures by K=5. The fitted results track that curve: skew-t K1 recovery ratios are 0.93 (K=5) $\rightarrow$ 0.49 (K=10) $\rightarrow$ 0.14 (K=25) $\rightarrow$ 0.073 (K=50) — the extended elbow, realised in MCMC. Two by-products: (a) the Brier anomaly resolution above; (b) a resolution ceiling — a higher-frequency mean component (3 cycles per axis) is *unrecoverable* on the 100-train/44-test geometry (its projection error *rises* with K), i.e. MRTS's usable resolution is capped by site density, and the six-bump surface sits just inside that cap.

5 The non-t spotlight quartet: settings 16, 19, 20, 21

Shared construction

- All four settings draw

$$Y_t(\boldsymbol{s}) \;=\; 10 \;+\; g(\boldsymbol{s}) \;+\; \sigma_g\,\varepsilon_t(\boldsymbol{s}), \qquad \varepsilon_t \overset{iid}{\sim} N(\mathbf{0},\, C_{\mathrm{exp}}), \quad \sigma_g = 4, \quad t = 1, \ldots, 50, \tag{6}$$

  so the marginal law is Gaussian and every t-based error model is misspecified. Methods, holdout and scoring are as in Section 1; the fitted grid is $K_{\mathrm{MRTS}} \in \{0, 30, 40, 50\}$ with $n = 3$ (the $n = 10$ / full-K upgrade of settings 16 and 19 is running).

- Each mean surface is the image of a raw field $\tilde g \in \mathbb{R}^{144}$ under the orthogonalise-and-standardise map

$$g \;=\; 11\,\frac{(I - P_X)\,\tilde g}{\widehat{\mathrm{sd}}\{(I - P_X)\tilde g\}}, \qquad P_X = X(X^\top X)^{-1}X^\top, \quad X = [\mathbf{1},\, s_1,\, s_2] \in \mathbb{R}^{144\times 3}, \tag{7}$$

  so that $g \perp \mathrm{col}(X)$ and $\widehat{\mathrm{sd}}(g) = 11$ exactly. Hence the $K = 0$ design spans *none* of g; composite raw terms are standardised to unit sd before summing, written $\mathrm{std}\{\cdot\}$ below.

- Recoverability is quantified before any MCMC by the noise-free out-of-sample projection error

$$e(K) \;=\; \frac{\|B_K^{\mathrm{te}}\hat\beta_K - g^{\mathrm{te}}\|_2}{\|g^{\mathrm{te}} - \bar g^{\mathrm{te}}\|_2}, \qquad \hat\beta_K \;=\; \arg\min_{\beta\in\mathbb{R}^K} \|B_K^{\mathrm{tr}}\beta - g^{\mathrm{tr}}\|_2, \tag{8}$$

  where B_K is the K-column MRTS basis built on the 100 training sites (superscripts: evaluated at the training / 44 test sites). Table 12 reports $e(K)$ together with the oracle Brier floor $R^\star$ recomputed under (6).

Setting 16 (`sigmoid_front`): the steepest recoverable front

- With $r(\boldsymbol{s}) = \|\boldsymbol{s} - (2,2)\|_2$ and width $w = 1$,

$$\tilde g_{16}(\boldsymbol{s}) \;=\; \tanh\bigl(w^{-1}\{r(\boldsymbol{s}) - 4.5\}\bigr). \tag{9}$$

- Since $\|\nabla \tilde g_{16}\| = w^{-1}\,\mathrm{sech}^2\bigl(w^{-1}\{r - 4.5\}\bigr) \le w^{-1}$, the transition is confined to the annulus $|r - 4.5| \lesssim w$, of width $\approx 2w = 2$ units. The width is calibrated to the site-density resolution cap of Section 5: $e(K)$ decays monotonically with no extrapolation blow-up (Table 12).

- Role: the sharpest *smooth* localised structure the 100-site training density supports — the fine-K stress test for mean recovery under a misspecified error law.

Setting 19 (`poly2`): the easy-end control

- With rescaled coordinates $(x_1, x_2) = \boldsymbol{s}/10$,

$$\tilde{g}_{19}(\boldsymbol{s}) \;=\; \mathrm{std}\big\{(x_1 - \tfrac{1}{2})^2 + (x_2 - \tfrac{1}{2})^2\big\} \;+\; \mathrm{std}\big\{(x_1 - \tfrac{1}{2})(x_2 - \tfrac{1}{2})\big\}. \tag{10}$$

- $\tilde{g}_{19}$ lies in the 6-dimensional space $\mathcal{P}_2$ of quadratic polynomials, and $(I - P_X)$ quotients out its affine part, so the target is the 3-dimensional pure-quadratic component span$\{s_1^2, s_1 s_2, s_2^2\}$ mod col(X). Accordingly $e(K)$ collapses fastest of all nine non-t surfaces ($0.23 \to 0.05 \to 0.015$ at $K = 5, 15, 50$).

- Role: control setting — a method that fails here fails for likelihood or sampler reasons, not basis-resolution reasons.

Setting 20 (`transform_mix`): boundary-concentrated curvature

- $$\tilde{g}_{20}(\boldsymbol{s}) \;=\; \mathrm{std}\{\log(1 + s_1)\} \;+\; \mathrm{std}\{e^{-s_1/2}\} \;+\; \mathrm{std}\{s_1/(1 + s_2)\}. \tag{11}$$

- Every term is monotone in s_1 with curvature maximal on the domain boundary, e.g. $\partial^2_{s_1} \log(1 + s_1) = -(1 + s_1)^{-2}$, largest at $s_1 = 0$ — exactly where the uniform site draw is sparsest. Hence the slowest $e(K)$ decay of the quartet (0.21 at $K = 50$).

- Role: the transformations an applied modeller actually writes down (log, exponential decay, ratio); MRTS must spend its resolution where the data density is lowest.

Setting 21 (`nonsmooth`): structural non-representability

- With $(x)_+ = \max(x, 0)$,

$$\tilde{g}_{21}(\boldsymbol{s}) \;=\; \mathrm{std}\{(s_1 - 5)_+\} \;+\; \mathrm{std}\{|s_2 - 4|\}. \tag{12}$$

- $\tilde{g}_{21} \in C^0 \setminus C^1$: its gradient jumps across the crease lines $\{s_1 = 5\}$ and $\{s_2 = 4\}$, while every MRTS column is a C^∞ thin-plate-spline eigenfunction. The projection error therefore *plateaus* instead of vanishing: $e(30) = 0.074$, $e(40) = 0.064$, $e(50) = 0.063$.

- Role: the mean-channel counterpart of the dependence settings 3–8 — structure that *no* K can fully represent, bounding the best MRTS fit away from the oracle at every K.

What the $n = 3$ pilots already show

- The unthresholded methods post consistent gains on all four settings: best energy ratios 0.93–0.97, variogram 0.90–0.97, Brier 0.88–0.94 (Tables 3–5, 7), with skew-t K5 on setting 20 the standout (0.71 energy, 0.60 Brier). Under a misspecified error law the mean channel finally moves all three scores in the same direction — unlike the skew-t-core settings, where the unthresholded methods were flat.

- The thresholded methods *invert* their skew-t-core role: on settings 16 and 21 their best-over-K energy ratios exceed 1 (respectively 1.53/1.72 and 1.05/1.21 for the K1/K5 variants), and on 19 the K1 variant loses too (1.34). Setting 20 is the lone exception (0.60 K1, 0.45 K5) — the only spotlight surface where the censored likelihood still profits from a mean basis.

- Yet the same thresholded fits *improve* their Brier scores everywhere (0.85–0.91), so their failure is a predictive-field (energy/variogram) phenomenon, not a tail-probability one. The realised gains also do not follow the $e(K)$ ordering, and every Brier ratio stays far above its floor $R^{\star}$ — both consistent with Finding 3's sharpness–extrapolation account.

6 In progress

1. **Non-t extension analysis (settings 13–21)**: the extension exists because all earlier settings share the skew-t core, so at large K the fitted skew-t is *correctly specified* — a confound for every "MRTS helps/hurts" conclusion. Its nine settings are now fitted and scored at pilot scale ($n = 3$; Tables 3–5 and 7; four settings introduced in Section 5), but the systematic write-up — in particular why the thresholded methods, the carriers of every skew-t-core gain, *lose* under several misspecified-error surfaces (energy ratios up to 2.9, settings 14–17) while the unthresholded methods post steady 4–7% gains — is still to be done.

2. n **upgrade**: settings 4–12 and the non-t block are $n = 3$ pilots; the headline contrasts (11, 12, and the disambiguation) merit $n = 10$. An automated fill script (`fill_16_19.ps1`: fit increment → re-score → verify overlap → delete fits) is running settings 16 and 19 first.

Table 2: Anatomy of the DGPs: marginal law of $Y_t(\boldsymbol{s})$, spatial dependence of the error process, and time-averaged true mean $\bar{\mu}(\boldsymbol{s})$ (its spatial sd in parentheses; per-dataset values for random means). $C_{\exp}(h) = 0.9\,e^{-h}$ is the shared stationary exponential correlation (Matérn $\nu = \frac{1}{2}$, $\rho = 1$, unit diagonal, i.e. a 10% nugget); "skew-t" is the t_3 core of (1) with $\lambda = 3$; $*$ denotes convolution with an independent additive field ($\sigma_{\text{add}} = 3$); f_1, f_2 are the two cosine bumps. For the non-t block (13–21) the error is $\sigma_g\,\varepsilon$ with $\varepsilon \sim N(0, C_{\exp})$, $\sigma_g = 4$ (so the marginal is Gaussian, sd 4), and every mean surface is orthogonalised against $[1, s_1, s_2]$ and rescaled to sd $= 11$.

ID	`surf_type`	marginal law	spatial dependence	mean structure $\bar{\mu}(\boldsymbol{s})$
1	invariant	skew-t	$C_{\exp}$, stationary	$10 + 5f_1 + 3f_2$ (3.7)
2	varying	skew-t	$C_{\exp}$, stationary	$10 + w_{1t}f_1 + w_{2t}f_2$, $(w_{1t}, w_{2t}) \sim N(0, \text{diag}(25, 9))$ per day; $\bar{\mu} \approx 10$ (0.4)
3	ns_dependence	skew-t $*$ Gaussian	$C_{\exp}$ + low-rank cosine $\sum_j \tau_j^2 \phi_j(\boldsymbol{s})\phi_j(\boldsymbol{s}')$; non-stationary, negative at long range	10 (RE mean-zero) (0.4)
4	ps_cov	skew-t (unchanged)	Paciorek–Schervish locally anisotropic kernel; scale, aspect and rotation each vary across the domain	10 exactly (0)
5	ps_add	skew-t $*$ Gaussian	$C_{\exp} + \sigma_{\text{add}}^2\, C_{\text{PS}}$, additive	10 (RE mean-zero) (0.4)
6	covdep_cov	skew-t (unchanged)	isotropic PS kernel with local range $\rho(\boldsymbol{s}) = e^{0.7 f_1(\boldsymbol{s})} \in [0.5, 2.0]$	10 exactly (0)
7	covdep_add	skew-t $*$ Gaussian	$C_{\exp} + \sigma_{\text{add}}^2\, C_{\rho(\boldsymbol{s})}$, additive	10 (RE mean-zero) (0.4)
8	fuentes_cov	skew-t (unchanged)	Fuentes 4-regime weighted GP mixture, ranges 0.4–4	10 exactly (0)
9	gh_marginal	skew-t $*$ std. Tukey g-and-h; $g(\boldsymbol{s}) = 0.8f_1$, $h(\boldsymbol{s}) = 0.12(1{+}f_2)$	$C_{\exp}$ in both components (transform perturbs correlation only mildly)	10 (standardised mean-zero) (0.4)
10	lambda_varying	skew-t with $\lambda(\boldsymbol{s}) = 3 + 3f_2 \in [0, 6]$	$C_{\exp}$, stationary	induced: $10 + \lambda(\boldsymbol{s})\,\bar{z}$, $\bar{z} = \frac{1}{T}\sum_t \sigma_t \lvert z_t \rvert$ (2.8)
11	invariant_strong	skew-t	$C_{\exp}$, stationary	$10 + 15f_1 + 9f_2$ (11.0)
12	mrts_multibump	skew-t	$C_{\exp}$, stationary	$10+$ six alternating-sign Gaussian bumps (width 1.6) (11.0)
	Non-t extension: Gaussian-GP errors, $\sigma_g = 4$, mean surface sd 11			
13	franke	Gaussian	$C_{\exp}$, stationary	$10+$ Franke's function (2 large + 1 medium bump, 1 fine dip) (11.0)
14	ridge_curved	Gaussian	$C_{\exp}$, stationary	$10+$ curved Gaussian ridge $\exp[-(s_2' - 0.5 - 0.25\sin 2\pi s_1')^2/2 \cdot 0.12^2]$ (11.0)
15	annulus	Gaussian	$C_{\exp}$, stationary	$10+$ Ricker ring $\psi((r - 3)/1.5)$, $r = \lVert \boldsymbol{s} - (5, 5) \rVert$ (11.0)
16	sigmoid_front	Gaussian	$C_{\exp}$, stationary	$10 + \tanh[(\lVert \boldsymbol{s} - (2, 2) \rVert - 4.5)/1.0]$ arc front (11.0)
17	gp_mean	Gaussian	$C_{\exp}$, stationary	$10+$ GP draw (Matérn ν=1.5, range 2), redrawn per dataset, fixed in t (11.0)
18	franke_sas	sinh-arcsinh (ϵ=0.8, δ=1), right-skewed	$C_{\exp}$, stationary	$10+$ Franke (identical to setting 13) (11.0)
19	poly2	Gaussian	$C_{\exp}$, stationary	$10+$ 2nd-order polynomial (bowl + saddle) (11.0)
20	transform_mix	Gaussian	$C_{\exp}$, stationary	$10 + \log(1{+}s_1) + e^{-s_1/2} + s_1/(1{+}s_2)$ (11.0)
21	nonsmooth	Gaussian	$C_{\exp}$, stationary	$10 + \text{hinge}(s_1{-}5) + \lvert s_2 - 4 \rvert$ (C^0 creases) (11.0)

Table 3: **Energy score**, best over $K > 0$ relative to the same method's $K = 0$ (minimising K in parentheses). Values < 1: MRTS helps; bold: $\bar{R} < 0.75$.

ID	non-stationarity	n	Gaussian	Skew-t K1	Sym-t K1 T	Skew-t K5	Sym-t K5 T
1	mean (2 bumps, sd 3.7)	10	0.994* (40)	0.980* (40)	**0.579*** (20)	0.983* (50)	**0.659*** (12)
3	dependence (low-rank)	10	1.000 (5)	1.000 (3)	0.993 (5)	1.000 (15)	0.911 (3)
4	dependence (PS, repl. C)	3	0.999* (5)	1.000 (30)	0.889* (40)	0.996* (3)	0.846 (15)
5	dependence (PS, additive)	3	0.999 (12)	0.999* (8)	0.953 (15)	0.998* (10)	0.955 (15)
6	dependence ($\rho(\boldsymbol{s})$, repl. C)	3	1.000 (10)	0.999 (10)	0.861* (20)	0.997* (12)	0.908 (50)
7	dependence ($\rho(\boldsymbol{s})$, additive)	3	1.000 (25)	0.999 (30)	0.985 (30)	0.995 (25)	0.926 (15)
8	dependence (Fuentes, repl. C)	3	1.000 (10)	1.000 (3)	0.848 (25)	0.998 (5)	**0.695** (30)
9	tails (g-and-h)	3	0.999* (5)	1.000 (10)	0.887 (5)	0.986 (25)	0.942 (5)
10	skewness ($\lambda(\boldsymbol{s})$)	3	0.993* (50)	1.000 (50)	0.975 (5)	0.995 (30)	0.938* (5)
11	mean (2 bumps, sd 11)	3	0.976* (50)	0.965* (50)	**0.199*** (20)	0.961* (50)	**0.267*** (20)
12	mean (6 bumps, sd 11)	3	0.960* (50)	0.953* (50)	**0.421*** (40)	0.952* (50)	0.846 (15)
Non-t extension (Gaussian-GP errors; skew-t is misspecified)							
13	mean, Franke (multiscale)	3	0.944* (50)	0.945* (50)	1.198 (50)	0.942* (50)	1.179 (40)
14	mean, curved ridge	3	0.956* (40)	0.956* (40)	2.263* (50)	0.953* (40)	2.938* (40)
15	mean, annulus (Ricker)	3	1.028* (50)	1.029* (50)	1.159 (30)	1.031* (50)	1.116* (40)
16	mean, sigmoid front	10	0.957* (20)	0.957* (50)	0.999 (3)	0.956* (50)	1.002 (3)
17	mean, GP draw	3	1.005 (50)	1.004 (50)	1.301 (40)	0.993 (50)	1.343* (40)
18	mean, Franke + sinh-arcsinh err.	3	0.941* (50)	0.942* (50)	0.851 (30)	0.950* (50)	0.975 (50)
19	mean, poly2 (bowl+saddle)	3	0.954* (30)	0.953* (50)	1.343 (40)	0.946* (50)	0.973 (40)
20	mean, transform mix	3	0.974* (50)	0.970* (50)	**0.597** (50)	**0.707*** (50)	**0.453*** (50)
21	mean, non-smooth (hinges)	3	0.949* (30)	0.947* (30)	1.052 (30)	0.927* (30)	1.214 (30)

Table 4: **Brier score** (mean over the 11 exceedance levels), best over $K > 0$ relative to own $K = 0$. Bold: $\bar{R} < 0.75$.

ID	non-stationarity	n	Gaussian	Skew-t K1	Sym-t K1 T	Skew-t K5	Sym-t K5 T
1	mean (2 bumps, sd 3.7)	10	1.000 (3)	0.978* (50)	0.993* (30)	0.984* (50)	0.971 (3)
3	dependence (low-rank)	10	1.000 (5)	0.999 (5)	1.000 (3)	0.996 (40)	0.993 (12)
4	dependence (PS, repl. C)	3	0.998 (50)	0.998* (30)	0.998* (30)	0.983* (50)	0.940* (20)
5	dependence (PS, additive)	3	1.000 (3)	0.999 (12)	1.000 (8)	0.990 (15)	0.973 (5)
6	dependence ($\rho(\boldsymbol{s})$, repl. C)	3	0.998 (50)	0.999* (5)	0.997* (40)	0.983* (12)	0.970 (10)
7	dependence ($\rho(\boldsymbol{s})$, additive)	3	1.000 (15)	0.999 (10)	0.998 (15)	0.990 (5)	0.968 (15)
8	dependence (Fuentes, repl. C)	3	1.000* (3)	1.000 (3)	0.999 (3)	0.997 (12)	0.962 (12)
9	tails (g-and-h)	3	1.000 (12)	1.000 (15)	0.998 (40)	0.996 (10)	0.967 (12)
10	skewness ($\lambda(\boldsymbol{s})$)	3	0.988 (50)	0.998 (50)	0.949* (50)	0.988 (30)	0.972 (30)
11	mean (2 bumps, sd 11)	3	1.000 (3)	1.000 (3)	0.952* (30)	1.004 (3)	0.997 (3)
12	mean (6 bumps, sd 11)	3	0.930 (50)	0.925* (50)	0.880* (50)	0.913* (50)	0.884* (10)
Non-t extension (Gaussian-GP errors; skew-t is misspecified)							
13	mean, Franke (multiscale)	3	0.974* (50)	0.972* (50)	0.956 (50)	0.950* (50)	0.936* (30)
14	mean, curved ridge	3	0.990 (50)	0.989* (50)	0.979 (40)	0.895* (50)	0.964* (40)
15	mean, annulus (Ricker)	3	1.143* (50)	1.142* (50)	0.874* (40)	1.144* (50)	0.924* (40)
16	mean, sigmoid front	10	0.950* (25)	0.949* (25)	0.916* (25)	0.935* (50)	0.912* (30)
17	mean, GP draw	3	0.984 (40)	0.982 (40)	0.913* (50)	0.971 (40)	0.867* (50)
18	mean, Franke + sinh-arcsinh err.	3	0.970* (50)	0.971* (50)	0.920* (30)	0.960* (50)	0.887* (30)
19	mean, poly2 (bowl+saddle)	3	0.931* (50)	0.927* (50)	0.907* (40)	0.896* (50)	0.885* (40)
20	mean, transform mix	3	0.941* (30)	0.936* (30)	0.851* (30)	**0.601*** (40)	**0.673*** (50)
21	mean, non-smooth (hinges)	3	0.918* (30)	0.917* (30)	0.896* (30)	0.875* (30)	0.852* (40)

Table 5: **Variogram score** ($p = 1/2$), best over $K > 0$ relative to own $K = 0$. Bold: $\bar{R} < 0.75$.

ID	non-stationarity	n	Gaussian	Skew-t K1	Sym-t K1 T	Skew-t K5	Sym-t K5 T
1	mean (2 bumps, sd 3.7)	10	0.994* (40)	0.975* (40)	**0.617*** (15)	0.981* (50)	**0.712** (12)
3	dependence (low-rank)	10	1.000 (5)	1.000 (3)	0.984 (5)	0.999 (25)	0.894 (3)
4	dependence (PS, repl. C)	3	0.999* (10)	1.000 (15)	0.988 (15)	0.997 (10)	0.990 (3)
5	dependence (PS, additive)	3	0.999* (12)	0.999* (8)	0.963 (50)	0.998 (15)	0.986 (15)
6	dependence ($\rho(\boldsymbol{s})$, repl. C)	3	0.999 (10)	0.999 (50)	0.920* (40)	0.997 (12)	0.984 (10)
7	dependence ($\rho(\boldsymbol{s})$, additive)	3	1.000 (10)	1.000 (30)	0.995 (3)	0.995 (5)	0.960 (15)
8	dependence (Fuentes, repl. C)	3	1.000 (8)	1.000 (3)	0.887 (40)	1.000 (3)	0.847 (5)
9	tails (g-and-h)	3	1.000 (3)	1.000 (5)	0.865 (5)	0.982 (25)	0.947 (5)
10	skewness ($\lambda(\boldsymbol{s})$)	3	0.990 (40)	0.999 (30)	1.000 (3)	0.993 (30)	0.905* (10)
11	mean (2 bumps, sd 11)	3	0.963* (50)	0.943* (50)	**0.201*** (10)	0.935* (50)	**0.244*** (20)
12	mean (6 bumps, sd 11)	3	0.938* (50)	0.911* (50)	**0.354*** (40)	0.915* (50)	0.909 (15)
Non-t extension (Gaussian-GP errors; skew-t is misspecified)							
13	mean, Franke (multiscale)	3	0.912* (50)	0.913* (50)	1.164* (30)	0.912* (50)	1.085 (30)
14	mean, curved ridge	3	0.891* (40)	0.891* (40)	2.092* (50)	0.913* (40)	2.571* (40)
15	mean, annulus (Ricker)	3	1.032* (50)	1.031* (50)	1.448* (30)	1.035* (50)	1.444* (40)
16	mean, sigmoid front	10	0.941* (20)	0.941* (50)	0.998 (3)	0.938* (20)	1.000 (3)
17	mean, GP draw	3	1.000 (50)	0.998 (50)	1.205 (40)	0.983 (50)	1.218 (40)
18	mean, Franke + sinh-arcsinh err.	3	0.905* (50)	0.906* (50)	**0.726*** (30)	0.919* (50)	0.854 (30)
19	mean, poly2 (bowl+saddle)	3	0.935* (30)	0.935* (50)	1.542 (40)	0.928* (50)	1.020 (40)
20	mean, transform mix	3	0.965* (50)	0.961* (50)	**0.562** (50)	**0.641*** (50)	**0.345*** (50)
21	mean, non-smooth (hinges)	3	0.934* (30)	0.932* (30)	1.055 (30)	0.904* (30)	1.209 (30)

Table 6: **Relative Brier score at every K_{MRTS}**, K1-knot methods (Gaussian, Skew-t K1, Sym-t K1 T), skew-t settings (vs the same method's own $K = 0$, mean over datasets; * marks $|\bar{R} - 1| > 2\,\widehat{\mathrm{SE}}$). A dash means that K was not fitted for that setting (settings 13–21 use only $K \in \{30, 40, 50\}$). Isolated large values of Sym-t K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.

ID	n	K=3	K=5	K=8	K=10	K=12	K=15	K=20	K=25	K=30	K=40	K=50
Gaussian												
1	10	1.000	1.016*	1.023*	1.022*	1.019*	1.016*	1.009*	1.008*	1.005	1.003	1.003
3	10	1.000	1.000	1.001*	1.001*	1.000	1.000	1.000	1.001	1.001	1.001*	1.002*
4	3	1.000	0.999*	0.999*	1.000	1.000	1.000	0.999	0.999	0.999	0.999	0.998
5	3	1.000	1.001	1.001	1.000	1.001*	1.000	1.001*	1.002*	1.002*	1.001*	1.004*
6	3	1.000	1.000	1.000	1.000	1.000	1.000	0.999	1.000	0.999	0.998*	0.998
7	3	1.000	1.000	1.000	1.000	1.000	1.000	1.002*	1.001	1.000	1.000	1.000
8	3	1.000*	1.001	1.001	1.001	1.000	1.000	1.002	1.002	1.002	1.003	1.004*
9	3	1.000*	1.000	1.000	1.000*	1.000	1.000	1.000	1.001	1.001	1.002	1.001
10	3	1.000	1.003	1.008	1.004	1.000	0.997	0.995	0.995	0.989	0.988	0.988
11	3	1.000	1.059*	1.078*	1.101*	1.114*	1.095*	1.069*	1.051*	1.022	1.001	1.000
12	3	1.000	1.003*	1.000	0.993	0.980	1.019	1.049	1.045	0.988	0.947	0.930
Skew-t K1												
1	10	1.000	0.996	1.006	1.006	1.001	0.996	0.987*	0.985*	0.981*	0.978*	0.978*
3	10	1.000	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
4	3	1.000*	1.000	0.999	1.000	1.000	1.000	0.999	0.999	0.998*	0.999	0.999
5	3	1.000	1.000	0.999	1.000	0.999	1.000	1.000	1.001	1.001	1.001	1.002
6	3	1.000	0.999*	0.999*	1.000	1.001*	0.999	1.000	1.000	1.000	0.999	1.000
7	3	1.000	1.000	1.001	0.999	1.000	1.000	1.001	1.000	1.000	1.001	1.001
8	3	1.000	1.000	1.001	1.000	1.001	1.001*	1.001	1.000	1.001	1.001*	1.002*
9	3	1.001	1.001	1.001*	1.001	1.001	1.000	1.001	1.001	1.001	1.001	1.001
10	3	0.999	1.003	1.008*	1.008	1.005	1.003	1.002	1.001	0.999	0.998	0.998
11	3	1.000	1.052*	1.074*	1.103*	1.163*	1.138*	1.110*	1.086*	1.048	1.017	1.014
12	3	1.000	1.005*	0.999	0.992	0.965	1.022	1.024	1.020	0.980	0.946	0.925*
Sym-t K1 T												
1	10	1.000	1.003	1.006	1.008*	1.003	1.001	0.996	0.994*	0.993*	0.993*	0.994*
3	10	1.000	1.000	1.001	1.001	1.001	1.001	1.001	1.001	1.002	1.002	1.004*
4	3	1.000	0.998*	0.999	0.999	1.000	0.999	1.000	0.999	0.998*	1.000	0.999
5	3	1.000*	1.000	1.000	1.000	1.000	1.001	1.000	1.001	1.002*	1.004	1.007
6	3	1.000	1.002*	1.000	1.000	1.001	0.999*	0.999	1.000	1.001	0.997*	1.000
7	3	1.000*	1.000	1.000	1.000	1.000	0.998	1.001	1.000	1.000	0.999	1.001
8	3	0.999	0.999	0.999	1.002	1.000	1.001	1.003	1.008	1.009	1.012	1.010*
9	3	1.000	1.001	1.001	1.000	1.001	1.001*	1.000	1.001*	1.002	0.998	1.000
10	3	0.999	0.993*	0.980*	0.985	0.975	0.970	0.963*	0.957*	0.949*	0.954*	0.949*
11	3	0.999	1.016	1.021	1.039*	1.008	0.982	0.977*	0.959*	0.952*	0.954*	0.959*
12	3	1.000	1.006*	0.997	0.907*	0.893*	0.899*	0.896*	0.918*	0.892*	0.896*	0.880*

Table 7: **Relative Brier score at every K_{MRTS}**, K1-knot methods (Gaussian, Skew-t K1, Sym-t K1 T), non-t extension (Gaussian-GP errors) (vs the same method's own $K = 0$, mean over datasets; * marks $|\bar{R} - 1| > 2\,\widehat{\text{SE}}$). A dash means that K was not fitted for that setting (settings 13–21 use only $K \in \{30, 40, 50\}$). Isolated large values of Sym-t K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.

ID	n	K=3	K=5	K=8	K=10	K=12	K=15	K=20	K=25	K=30	K=40	K=50
Gaussian												
13	3	–	–	–	–	–	–	–	–	1.025	0.999	0.974*
14	3	–	–	–	–	–	–	–	–	1.052*	1.059*	0.990
15	3	–	–	–	–	–	–	–	–	1.182*	1.172*	1.143*
16	10	1.000	0.991*	0.992	0.996	1.009	0.986*	0.960*	0.950*	0.955*	0.952*	0.951*
17	3	–	–	–	–	–	–	–	–	1.008	0.984	0.991
18	3	–	–	–	–	–	–	–	–	1.020	0.997	0.970*
19	3	–	–	–	–	–	–	–	–	0.935*	0.935*	0.931*
20	3	–	–	–	–	–	–	–	–	0.941*	0.970*	0.967*
21	3	–	–	–	–	–	–	–	–	0.918*	0.921*	0.923*
Skew-t K1												
13	3	–	–	–	–	–	–	–	–	1.024	0.998	0.972*
14	3	–	–	–	–	–	–	–	–	1.055*	1.061*	0.989*
15	3	–	–	–	–	–	–	–	–	1.180*	1.169*	1.142*
16	10	1.000	0.990*	0.992	0.997	1.010	0.986*	0.958*	0.949*	0.954*	0.951*	0.951*
17	3	–	–	–	–	–	–	–	–	1.006	0.982	0.988
18	3	–	–	–	–	–	–	–	–	1.022	0.999	0.971*
19	3	–	–	–	–	–	–	–	–	0.932*	0.932*	0.927*
20	3	–	–	–	–	–	–	–	–	0.936*	0.965*	0.961*
21	3	–	–	–	–	–	–	–	–	0.917*	0.917*	0.920*
Sym-t K1 T												
13	3	–	–	–	–	–	–	–	–	0.960	0.989	0.956
14	3	–	–	–	–	–	–	–	–	1.079*	0.979	0.980
15	3	–	–	–	–	–	–	–	–	0.945*	0.874*	0.952
16	10	0.999	0.995	0.973*	0.975*	0.932*	0.937*	0.921*	0.916*	0.916*	0.924*	0.925*
17	3	–	–	–	–	–	–	–	–	0.930	0.917*	0.913*
18	3	–	–	–	–	–	–	–	–	0.920*	0.931*	0.946
19	3	–	–	–	–	–	–	–	–	0.912*	0.907*	0.930*
20	3	–	–	–	–	–	–	–	–	0.851*	0.899*	0.902
21	3	–	–	–	–	–	–	–	–	0.896*	0.907*	0.909*

Table 8: **Relative Brier score at every K_{MRTS}**, K5-knot methods (Skew-t K5, Sym-t K5 T), all settings (vs the same method's own $K = 0$, mean over datasets; * marks $|\bar{R} - 1| > 2\,\widehat{\mathrm{SE}}$). A dash means that K was not fitted for that setting (settings 13–21 use only $K \in \{30, 40, 50\}$). Isolated large values of Sym-t K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.

ID	n	K=3	K=5	K=8	K=10	K=12	K=15	K=20	K=25	K=30	K=40	K=50
Skew-t K5												
1	10	0.995	1.004	1.015*	1.008	1.010	1.011*	0.993	0.994	0.991	0.992	0.984*
3	10	0.999	1.004	1.006	0.998	1.001	0.997	1.000	0.998	1.003	0.996	1.000
4	3	0.989*	0.993	0.988	0.994	0.992	0.994	0.987	0.985*	0.994	0.988	0.983*
5	3	0.996	0.994	0.992*	0.992	0.994	0.990	0.993	0.997	0.996	0.993	0.997
6	3	0.998	1.001	1.000	0.996	0.983*	0.996	0.986*	0.989	1.002	0.988	1.005
7	3	0.993	0.990	0.997	0.994	0.994	0.994	0.994	0.993	0.995	0.995	1.002
8	3	0.999	0.999	1.007	1.011*	0.997	1.002	1.015*	1.007*	1.005	1.012*	1.008
9	3	1.005	1.007	0.999	0.996	1.007	1.001	1.003	1.003	1.001	1.005	1.003
10	3	0.999	1.002*	1.006*	1.008	1.004	1.005	1.002	0.994	0.988	0.998	0.996
11	3	1.004	1.088*	1.096*	1.123*	1.184*	1.155*	1.142*	1.112	1.073	1.038	1.036
12	3	0.992	1.034	0.995	1.004	0.962	0.976	1.017	1.022	0.965	0.927*	0.913*
13	3	–	–	–	–	–	–	–	–	1.001	0.977*	0.950*
14	3	–	–	–	–	–	–	–	–	0.983*	0.981*	0.895*
15	3	–	–	–	–	–	–	–	–	1.177*	1.168*	1.144*
16	10	0.995*	0.977*	0.975*	0.985*	0.996	0.971*	0.945*	0.935*	0.941*	0.936*	0.935*
17	3	–	–	–	–	–	–	–	–	0.996	0.971	0.980
18	3	–	–	–	–	–	–	–	–	1.018*	0.992	0.960*
19	3	–	–	–	–	–	–	–	–	0.900*	0.899*	0.896*
20	3	–	–	–	–	–	–	–	–	0.618*	0.601*	0.601*
21	3	–	–	–	–	–	–	–	–	0.875*	0.877*	0.879*
Sym-t K5 T												
1	10	0.971	1.057	1.073	1.097*	1.055	1.023	1.050	1.085	1.131	1.073	1.165*
3	10	0.997	1.011	1.001	1.009	0.993	1.013	1.068	1.066	1.007	1.124	1.160
4	3	0.981*	0.980	0.999	0.998	1.022	0.994	0.940*	0.995	1.056	0.993	0.961
5	3	1.021	0.973	0.975	0.974	0.987	0.980	0.977	0.973	1.023	1.009	1.026
6	3	0.980	0.978	0.994	0.970	0.988	1.020	0.997	1.028	1.033*	0.998	1.045*
7	3	0.981	0.990	0.987	0.972	0.993	0.968	1.011	0.993	1.029	1.017	1.021
8	3	0.967	1.005	0.990	0.999	0.962	1.019	1.063*	1.050	1.084	0.988	1.063
9	3	0.992	0.982	1.015	0.991	0.967	1.008	0.998	1.040*	1.029	1.045	0.974
10	3	0.990	1.005	0.996	1.027	1.004	0.994	0.988	1.010	0.972	0.977	0.972
11	3	0.997	1.078	1.182*	1.114*	1.163*	1.120*	1.104	1.133	1.030	1.086	1.063
12	3	0.994*	0.998	0.999	0.884*	0.903*	0.966	0.917*	1.321	1.012	1.061	1.587
13	3	–	–	–	–	–	–	–	–	0.936*	0.947	0.956
14	3	–	–	–	–	–	–	–	–	1.105*	0.964*	1.027
15	3	–	–	–	–	–	–	–	–	0.924*	0.924*	0.935*
16	10	1.000	0.992	0.967*	0.971*	0.922*	0.927*	0.918*	0.913*	0.912*	0.919*	0.949*
17	3	–	–	–	–	–	–	–	–	0.907*	0.903*	0.867*
18	3	–	–	–	–	–	–	–	–	0.887*	0.962	0.893*
19	3	–	–	–	–	–	–	–	–	0.904*	0.885*	1.067
20	3	–	–	–	–	–	–	–	–	0.730*	0.784*	0.673*
21	3	–	–	–	–	–	–	–	–	0.853*	0.852*	0.854*

Table 9: Oracle Brier floor $R^\star$ per setting (`oracle_brier_ceiling.R`; Monte Carlo, 3 datasets).

setting	sd($\bar{\mu}$)	$R^\star$	verdict
1 invariant	3.68	0.856	weak headroom (14%)
2 varying	0.42	1.000	dead end
3 ns_dependence	0.41	0.999	dead end
4 ps_cov	0.00	1.000	dead end
5 ps_add	0.39	0.999	dead end
6 covdep_cov	0.00	1.000	dead end
7 covdep_add	0.41	1.000	dead end
8 fuentes_cov	0.00	1.000	dead end
9 gh_marginal	0.43	1.000	dead end
10 lambda_varying	2.80	**1.120**	**harmful**
11 invariant_strong	11.04	0.439	positive control
12 mrts_multibump	11.00	0.495	positive control

Table 10: Setting 11, relative score vs K_{MRTS} ($n = 3$): the thresholded symmetric-t methods against the unthresholded skew-t.

score	method	K=3	K=5	K=8	K=10	K=12	K=15	K=20	K=25	K=30	K=40	K=50
energy	Skew-t K1 (no thr.)	0.999	1.013	1.022	1.027	1.023	1.017	0.996	0.984	0.978	0.966	0.965
	Sym-t K1 (thr.)	0.990	**0.263**	0.266	0.229	0.201	0.244	**0.199**	0.256	0.233	0.268	0.305
	Sym-t K5 (thr.)	1.001	0.317	0.539	0.374	0.342	0.462	**0.267**	0.333	0.359	0.906	0.860
Brier	Skew-t K1 (no thr.)	1.000	1.052	1.074	1.103	**1.163**	1.138	1.110	1.086	1.048	1.017	1.014
	Sym-t K1 (thr.)	0.999	1.016	1.021	1.039	1.008	0.982	0.977	0.959	0.952	0.954	0.959
	Sym-t K5 (thr.)	0.997	1.078	1.182	1.114	1.163	1.120	1.104	1.133	1.030	1.086	1.063

Table 11: Disambiguation on setting 1 ($n = 10$): relative energy score vs K_{MRTS}, and the *absolute* energy score at $K = 0$. Method pairs $(3, 11)$ and $(5, 12)$ differ *only* in the POT threshold.

	method	K=5	K=10	K=15	K=20	K=25	abs. ES at K=0
3	Sym-t K1, thr. $q(0.80)$	0.630	0.617	0.589	**0.579**	0.596	21.8
11	Sym-t K1, no threshold	0.996	0.997	0.991	0.990	0.989	8.5
5	Sym-t K5, thr. $q(0.80)$	0.688	1.113	0.773	0.718	**0.697**	137.1
12	Sym-t K5, no threshold	0.994	1.008	0.993	0.997	0.997	9.3
2	Skew-t K1 (reference)	0.994	0.994	0.985	0.983	0.982	8.5

Table 12: The spotlight quartet: noise-free projection error $e(K)$ of (8) and oracle Brier floor $R^\star$ (Monte Carlo, Gaussian error, $\sigma_g = 4$; `nont_precheck.R`).

ID	`surf_type`	$e(5)$	$e(15)$	$e(30)$	$e(50)$	$K \to \infty$ behaviour	$R^\star$
16	sigmoid_front	0.591	0.250	0.131	0.094	$\downarrow$, elbow at $K \approx 40$–50	0.559
19	poly2	0.226	0.054	0.030	0.015	$\downarrow$, saturated by $K \approx 12$	0.286
20	transform_mix	0.895	0.506	0.272	0.214	$\downarrow$, slowest decay	0.396
21	nonsmooth	0.469	0.159	0.074	0.063	plateau $\approx 0.063 > 0$	0.431